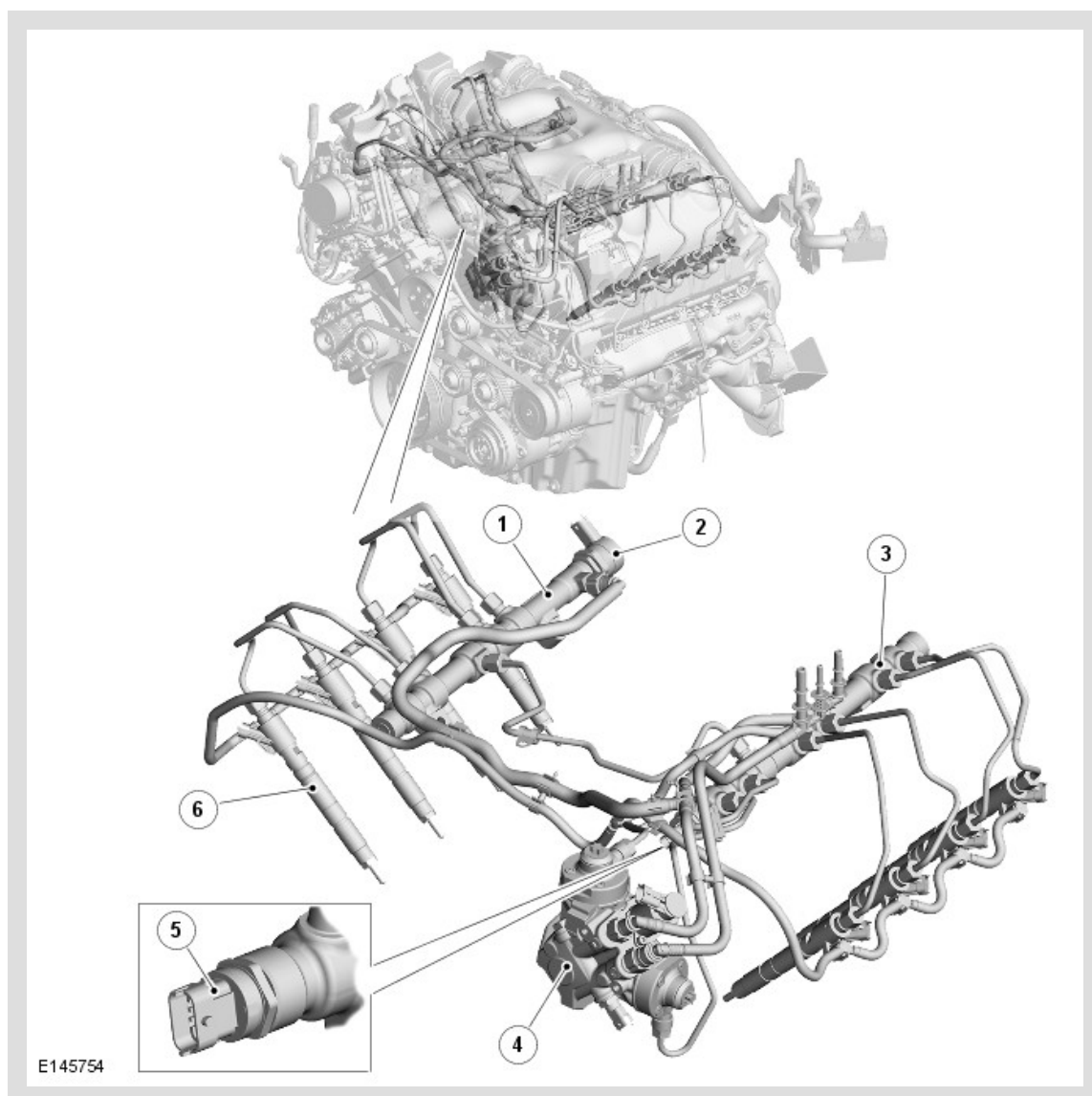


PUBLISHED: 07-AUG-2012

2015.0 RANGE ROVER (LG), 303-04

FUEL CHARGING AND CONTROLS - TDV8 4.4L DIESEL

COMPONENT LOCATION



ITEM	DESCRIPTION
1	Right fuel rail
2	Fuel pressure control valve
3	Left fuel rail
4	High pressure fuel pump
5	Fuel rail pressure sensor
6	Fuel injector (8 off)

OVERVIEW

The TDV8 4.4L diesel engine is equipped with a HP (high pressure) common rail fuel injection system. With this fuel injection process, a HP fuel pump delivers a uniform level of pressure to the shared fuel lines (the common rails), which serve all 8 fuel injectors. Pressure is controlled to the optimum level for smooth operation.

The common rail system supports a pre and pilot injection depending on engine operating conditions, which reduces combustion noise levels, more commonly referred to as 'diesel knock'.

Fuel injection pressure is generated independently of engine speed and fuel injection events.

The fuel injection timing and volume are calculated by the ECM (engine control module) , which then energizes the appropriate piezo actuated injector.



CAUTION:

After engine shut down the common rails are maintained at a high pressure ready for the next engine start. Pressure will only fall to zero as you start to release the first pipe union, so care must be taken to prevent fuel leakage and contain any fuel losses until the fuel pressure has dissipated.

The common rail fuel injection system has the following features:

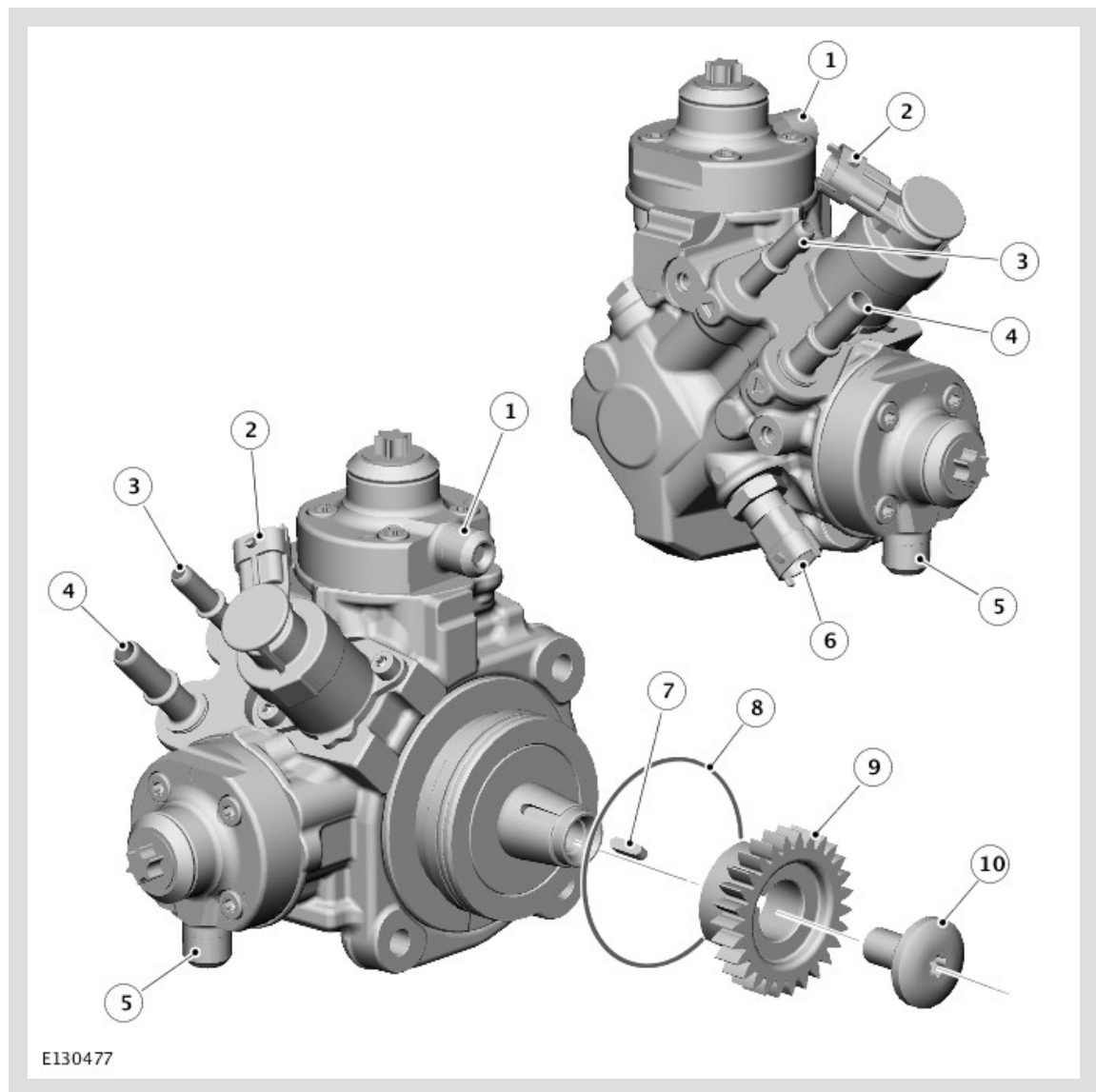
- High fuel injection pressures of up to 2000 bar (29007 lbf/in²) for greater atomisation of fuel (increasing performance and lowering emissions)
- Variable injection to optimize combustion in all engine operating conditions
- Low tolerances and high precision throughout the life of the system.

The system features the following components:

- HP fuel pump
- Fuel rails
- HP and LP (low pressure) fuel pipes
- Fuel injectors.

DESCRIPTION

HIGH PRESSURE FUEL PUMP



ITEM	DESCRIPTION
1	High pressure fuel outlet to left fuel rail
2	Fuel metering valve
3	Low pressure fuel return connection
4	Low pressure fuel supply connection
5	High pressure fuel outlet to left fuel rail
6	Fuel temperature sensor
7	Woodruff key
8	O-ring seal
9	Fuel injection pump drive gear
10	Drive flange bolt

The HP fuel pump is a 2-piston (90 degree apart) radial plunger pump with a HP displacement of 1.0 cc and the ability to produce a maximum pressure of 2000 bar (29007 lbf/in²).

The pump is driven from the left intake camshaft via a drive gear and must be timed to the engine.

For additional information, refer to: [Fuel Pump](#) (303-04F Fuel Charging and Controls - TDV8 4.4L Diesel, Removal and Installation).

The required supply pressure to the pump is 4.5 bar (65.3 lbf/in²) to 7.0 bar (101.5 lbf/in²). The return pressure is -0.3 bar to +0.8 bar.

The pump is sized to deliver sufficient fuel to the fuel rails for all engine operating conditions.

The HP fuel pump consists of the following components:

- Fuel metering valve
- HP pumping elements (2 off)
- Fuel temperature sensor.

The fuel metering valve is a variable position solenoid valve electronically controlled by the ECM . The fuel metering valve is located between the fuel inlet and the HP pumping elements. The fuel metering valve regulates the amount of fuel that is delivered to the fuel rails via the pumping elements. When there is no signal to the fuel metering valve the valve is closed, therefore there is no fuel delivery.

The 2 HP pumping elements each have a HP outlet that connects to the left fuel rail via HP pipes.

The HP fuel pump can supply up to 2000 bar fuel pressure continuously with a pump speed which is the same as engine speed. It is calibrated to deliver fuel pressure dependant upon engine speed and load and is always under full control.

When the HP fuel pump is rotated, fuel pressure is created when the fuel metering valve is open and the fuel pressure control valve in the right common rail is closed. The fuel metering valve and fuel pressure control valve are variable position to allow variable fuel delivery and pressure control.

When the ECM actuates the fuel injectors, the pressure drop in the fuel rails is offset by additional fuel being delivered by the HP fuel pump.

FUEL METERING VALVE

The fuel metering valve is fixed directly to the HP fuel pump.

The fuel metering valve regulates the fuel supply (and hence the quantity of fuel) from the internal transfer pump to the HP pumping elements, depending on the fuel pressure required in the fuel rails. This makes it possible to match the delivery of the HP fuel pump to the requirements of the engine from the LP side. The quantity of fuel flowing back to the main fuel supply line is kept to a minimum. In addition, this adjustment reduces the power consumption of the HP fuel pump, improving the efficiency of the engine.



NOTE:

The fuel metering valve default is closed without electrical supply. An open circuit connector will prevent the engine from running.

The fuel metering valve forms part of the HP fuel pump and cannot be replaced as a separate component in service.

FUEL PRESSURE CONTROL VALVE

The fuel pressure control valve is located on the right fuel rail. It governs the fuel pressure at the HP outlets of the HP fuel pump and thus the fuel pressure within the rails. In addition, the fuel pressure control valve dampens any fluctuations in pressure, which occur during the delivery of fuel through the HP fuel pump and through the injection process.

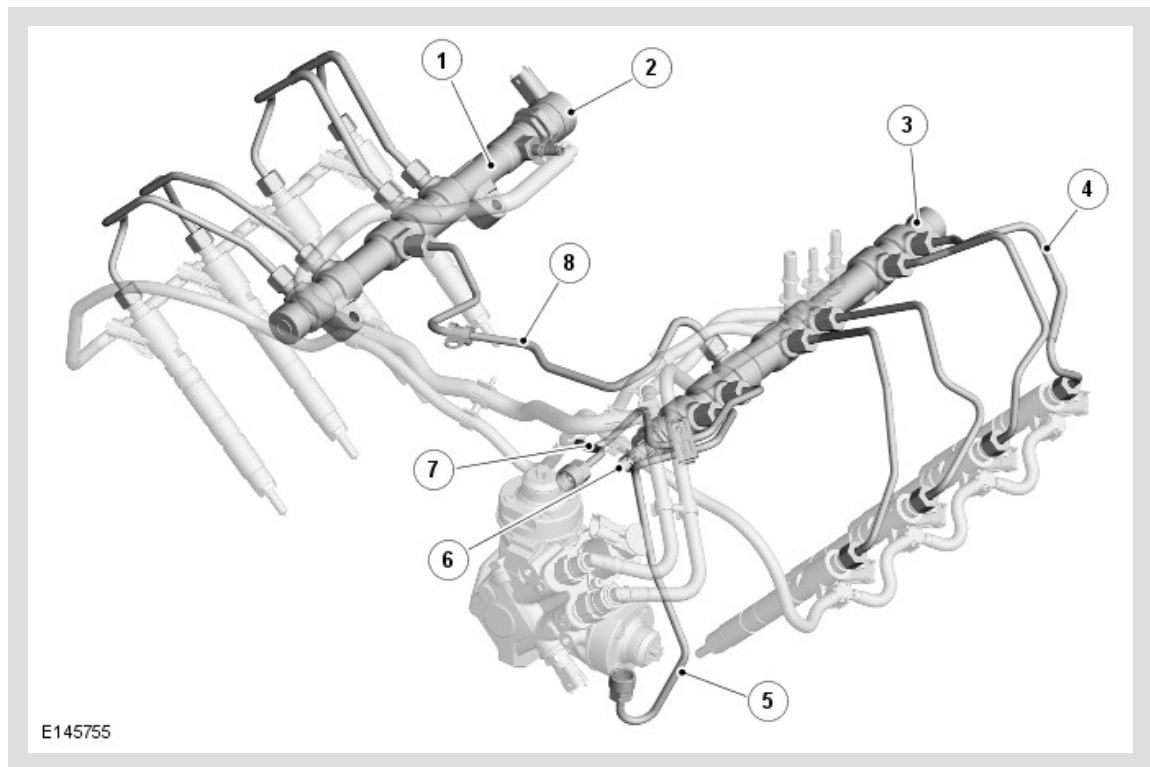
The fuel pressure control valve ensures that optimum pressure exists in the fuel rails for every operating condition of the engine.

The fuel pressure control valve is an electro-magnetically operated valve with spring support.

The electrical current supplied by the ECM through the solenoid attracts the armature to the coil. This forces the ball valve onto its seat via the control pin. The entire armature is coated with fuel for lubrication and cooling.

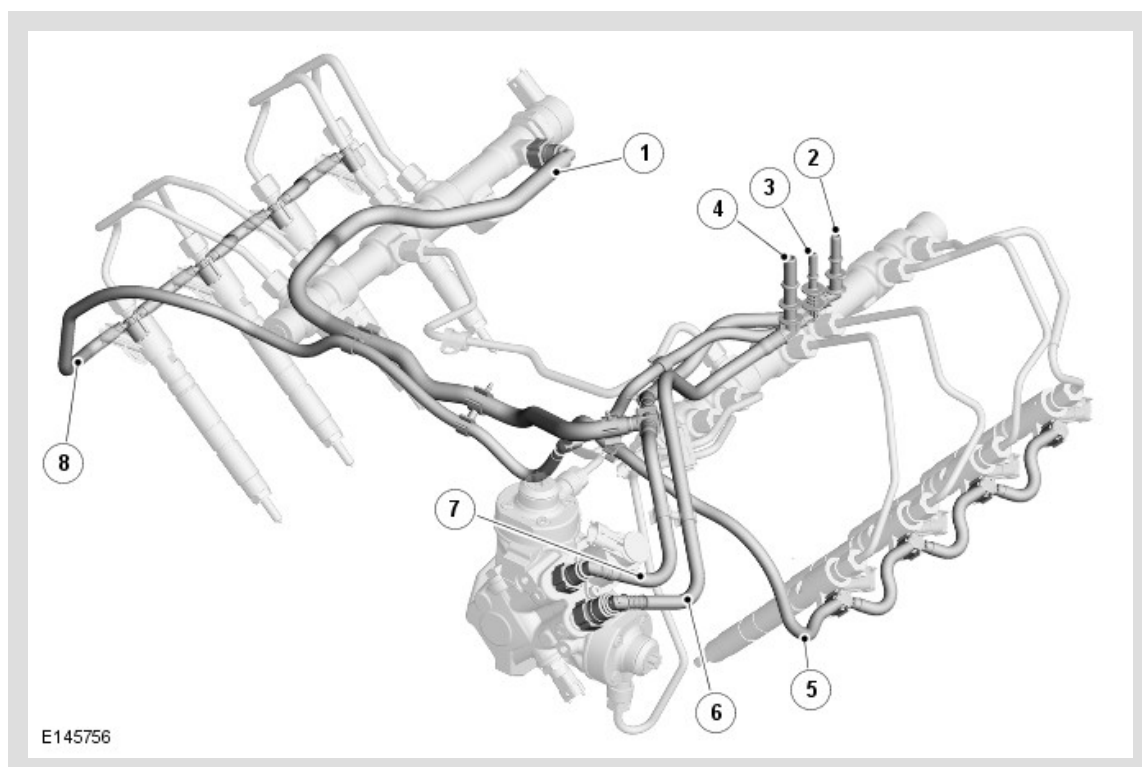
FUEL RAILS

Fuel Rails and High Pressure Fuel Lines



ITEM	DESCRIPTION
1	Right fuel rail
2	Fuel pressure control valve
3	Left fuel rail
4	Supply pipe to fuel injector (8 off)
5	Supply pipe to left fuel rail
6	Fuel rail pressure sensor
7	Supply pipe to left fuel rail
8	Fuel rail balance pipe

Low Pressure Fuel Lines



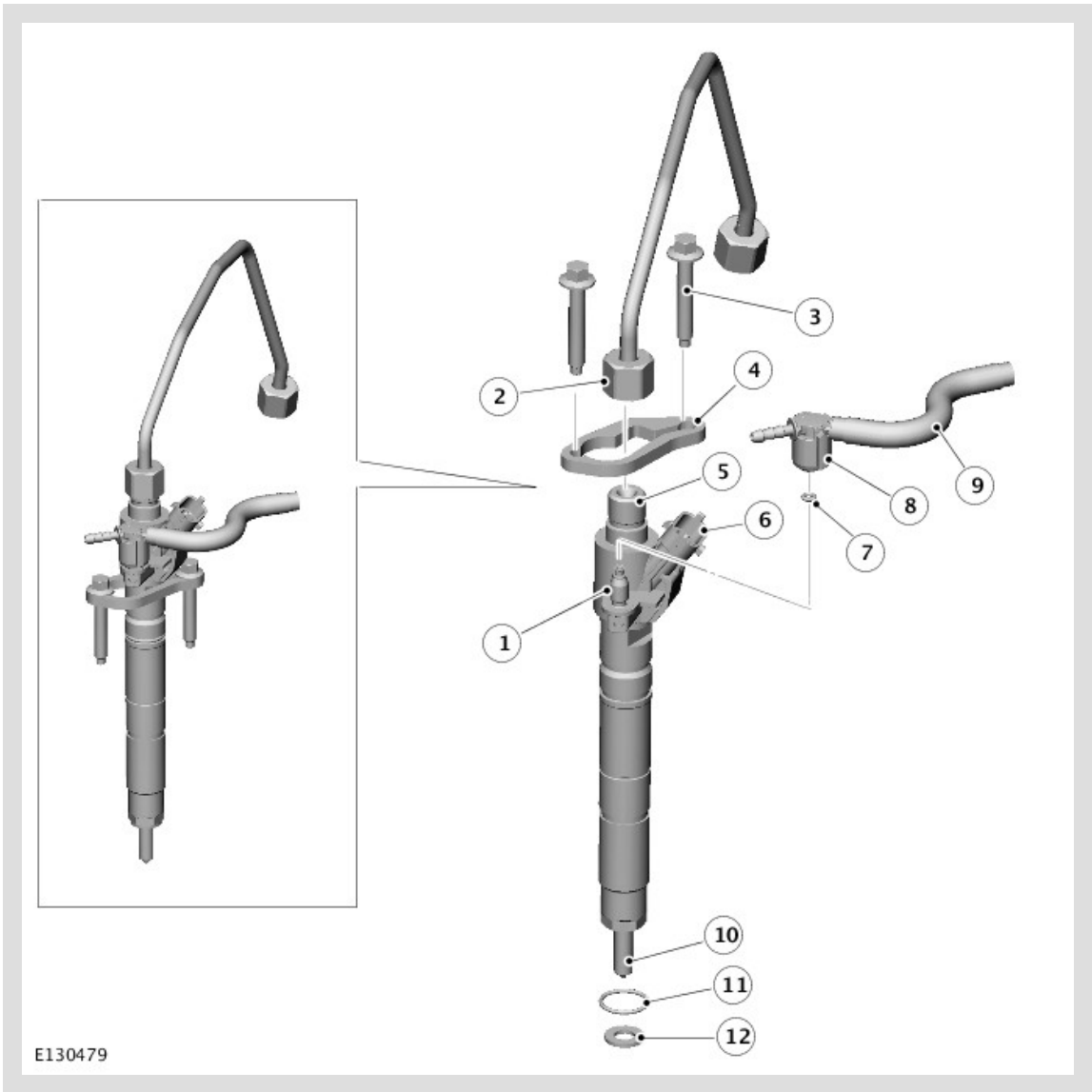
ITEM	DESCRIPTION
1	Fuel pressure control valve fuel return tube
2	Fuel return connection - fuel pressure control valve and high pressure fuel pump to fuel cooler
3	Fuel return connection - fuel injector leak-back to fuel filter inlet tube
4	Fuel supply connection - fuel filter to high pressure fuel pump
5	Left bank fuel injector leak-back tube
6	High pressure fuel pump fuel supply tube
7	High pressure fuel pump fuel return tube
8	Right bank fuel injector leak-back tube

The fuel rails store the fuel at high pressure and prevent pressure fluctuations in the HP system.

All HP pipes have an internal diameter of 3.0 mm. Total rail volume is 33 cc.

The left rail is fitted with a fuel pressure sensor and the right rail is fitted with a fuel pressure control valve. Both the pressure sensor and the fuel pressure control valve are connected to and controlled by the ECM .

FUEL INJECTORS



ITEM	DESCRIPTION
1	Leak-back connection
2	High pressure supply pipe from fuel rail
3	Bolt (2 off per injector)
4	Injector clamp
5	High pressure connection
6	Electrical harness connection

7	O-ring seal - leak back coupling
8	Injector leak-back coupling
9	Fuel leak-back tube
10	Nozzle spray holes
11	O-ring seal
12	Injector nozzle washer



CAUTION:

Each injection event is controlled by a charge and discharge cycle allowing energy to dissipate in, and recover from, the injector. Never disconnect the wiring connector when the vehicle is running. The injector may remain open thus causing engine damage.

The fuel injectors are operated directly by the ECM for fuel metering (start of injection and quantity of fuel injected).

The operating components of the piezo fuel injectors are:

- The piezo actuator
- The injector body containing the hydraulic servo system
- The fuel injector nozzle.

Each injector has a code which must be programmed into the ECM to identify which cylinder it is associated with. Therefore it is important that the injectors must be replaced into the correct cylinders.

OPERATION

ENGINE STARTING

During starting, the fuel rail pressure must be at least 150 bar (2175 lbf/in²). Should the pressure be below this figure, the injectors will not operate, resulting in the vehicle not starting.

ENGINE STOPPING

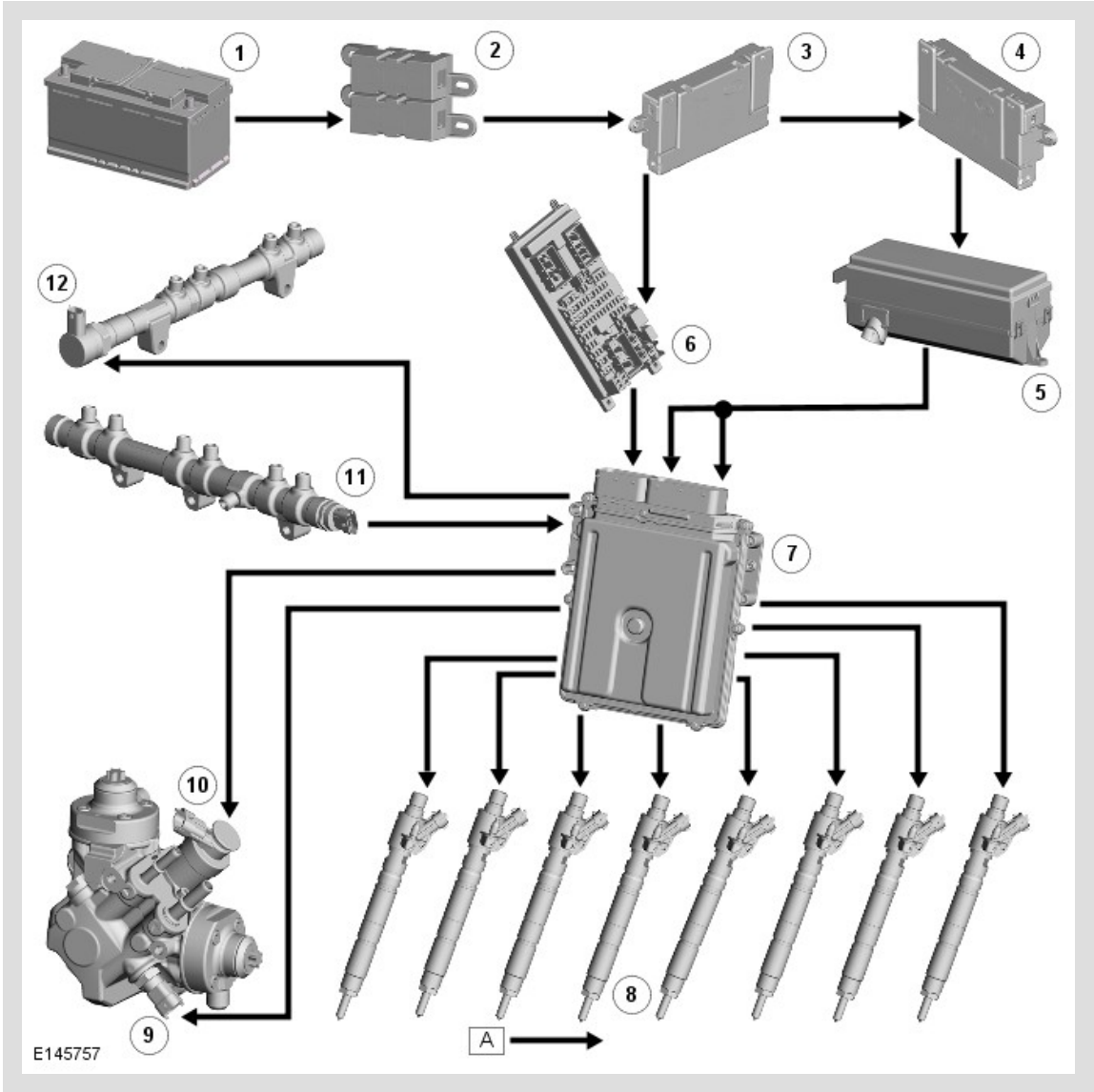
To stop the engine the ECM stops energizing the actuators in the fuel injectors, therefore, no fuel is injected and the engine speed drops to zero.

For additional information, refer to: [Electronic Engine Controls](#) (303-14A Electronic

HP FUEL PUMP

When the HP fuel pump is rotated, pressure is created when the volume control valve is open and the pressure control valve is closed. Both valves are electronically controlled by the ECM to allow variable fuel delivery and pressure control. When the ECM actuates the fuel injectors, the fuel rail pressure drop is off-set by additional fuel being delivered to the fuel rails by the pressure control valve. The fuel pressure in the system is reduced within a few seconds after the engine has stopped as the pressure control valve no longer has the holding current it requires, and therefore opens. No residual pressure remains in the system and the fuel is released into the LP fuel return line through the open pressure control valve.

CONTROL DIAGRAM



ITEM	DESCRIPTION
1	Battery

2	Battery junction box 2
3	Battery junction box
4	Auxiliary junction box
5	Engine junction box
6	Central junction box
7	Engine control module
8	Fuel injector (8 off)
9	Fuel temperature sensor
10	Fuel volume control valve
11	Fuel rail pressure sensor
12	Fuel pressure control valve